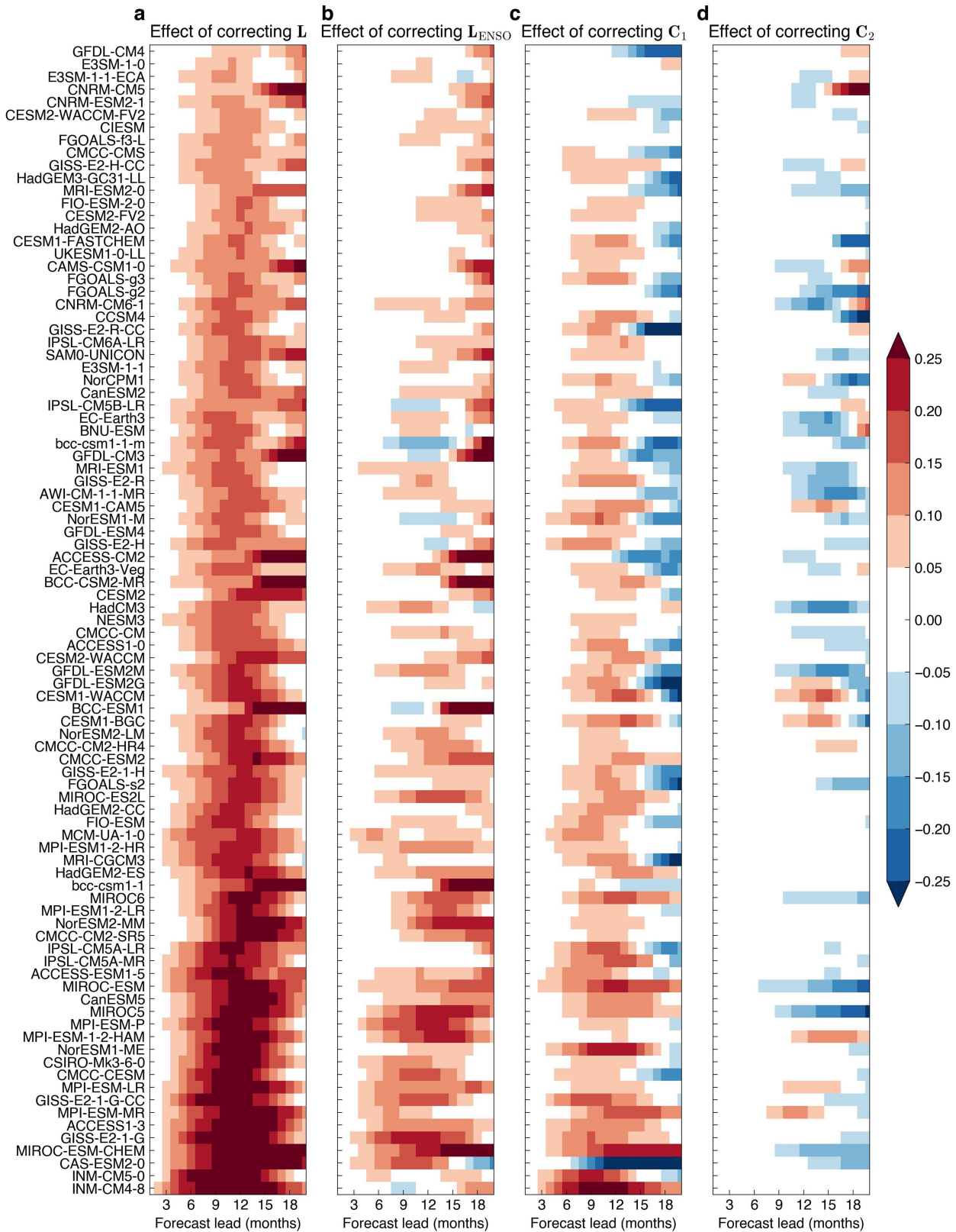




Extended Data Fig. 7 | Impacts on ENSO forecast skill of correcting biases in the XRO parameters fitted to individual CMIP simulations. Shown is the difference of the all-months correlation skill for the Niño3.4 SSTA index, between the corrected-parameter forecast experiment and the XRO^m experiment trained solely on CMIP model outputs. (a) Effect of correcting linear operators ($XRO_L^m - XRO^m$), (b) effect of correcting ENSO internal linear dynamics ($XRO_L^m - XRO^m$), (c) effect of correcting remote climate mode feedbacks onto ENSO ($XRO_{C_1}^m - XRO^m$), and (d) effect of correcting ENSO

teleconnections to remote climate modes ($XRO_{C_2}^m - XRO^m$). The model is sorted by the averaged correlation skill of the XRO^m forecast at 6–15 lead months. Reforecasts using the XRO trained on global climate model output show that correcting CGCMs' dynamical biases in ENSO and climate mode interactions lead to more skillful ENSO forecasts. Most important is correcting ENSO biases (which improves skill at longest lead-times), followed by correcting the remote climate mode impact on ENSO (which improves skill at intermediate leads). Less skill is gained by improving ENSO's teleconnection to the remote modes.